

P03 - Scalable Lakehouse Pipeline

Shipment events | explicit schema -> distributed transforms -> Parquet/Delta reasoning -> lakehouse -> Fabric mapping

Business/domain

FleetRoute shipment events

How this book works

This book has exactly 13 phase pages. Each phase has one artifact/evidence contract and one protected rubric page. The small teaching pattern is not the final solution; apply the reasoning to the supplied project workspace yourself.

Project control before Phase 1

- Keep protected Review and sealed failure/incident packs closed until the named phase.
- Create a real project branch/working copy if using Git.
- Record tool/runtime versions only after observing them.
- Do not delete a bad first attempt.
- Use the exact phase evidence ID and path; this keeps Mentor/book/evidence routing one-to-one.

Phase map

P03-01 - Read the scalable-lakehouse brief and define scale risks

P03-02 - Profile event schema, grain, keys and timestamps

P03-03 - Create Spark environment and explicit schemas

P03-04 - Build deterministic distributed transformations

P03-05 - Join reference data and validate cardinality

P03-06 - Reason about partitions, shuffle and skew

P03-07 - Write Parquet with justified partition strategy

P03-08 - Apply Delta-style incremental merge/idempotency

P03-09 - Define Bronze/Silver/Gold lakehouse contracts

P03-10 - Handle schema evolution and supplied failure cases

P03-11 - Inspect plans/performance and prove equivalent results

P03-12 - Map the design to Fabric/cloud deployment

P03-13 - Clean rerun, reconcile outputs and defend trade-offs

P03-01 - Read the scalable-lakehouse brief and define scale risks

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete read the scalable-lakehouse brief and define scale risks and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Scale risk is not synonymous with row count. Identify data shape, skew, file counts, join patterns, freshness and growth assumptions first.

TINY PATTERN - NOT YOUR FINAL SOLUTION

A million evenly distributed rows can be easier than fewer rows dominated by one hot key. Write the risk you would measure.

BUILD THIS ARTIFACT

business_questions.md + assumptions.md

VALIDATE

Read outputs back, verify schema/counts/keys, and prove partition/schema-evolution behavior on a changed input.

CHANGED / FAILURE TEST

PHASE CHECK P03-01: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-01-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-01/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-02.

P03-02 - Profile event schema, grain, keys and timestamps

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete profile event schema, grain, keys and timestamps and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Event grain includes identity and event time. Late/out-of-order events mean "latest" must be defined, not assumed from file order.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Profile event_id, shipment_id, event_time validity, repeated shipment_id and schema drift separately.

BUILD THIS ARTIFACT

source_profile/ + grain_key_defects.md

VALIDATE

Record row counts, grain, keys, null/duplicate/schema defects; independently sample-check at least one finding.

CHANGED / FAILURE TEST

Add/remove/change one field according to supplied compatibility rule; quarantine/fail safely rather than silently corrupt.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-02-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-02/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-03.

P03-03 - Create Spark environment and explicit schemas

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete create spark environment and explicit schemas and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Explicit schema turns ingestion assumptions into code. Record Spark/Java/Python versions before diagnosing data logic.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Define event_id/string, shipment_id/string, event_time/timestamp, weight/double; malformed records must remain visible.

BUILD THIS ARTIFACT

Spark job/notebook + explain plan + partition/performance evidence

VALIDATE

Compare output to small trusted control result; inspect plan/partitions; justify performance choice without changing correctness.

CHANGED / FAILURE TEST

Add/remove/change one field according to supplied compatibility rule; quarantine/fail safely rather than silently corrupt.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-03-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-03/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-04.

P03-04 - Build deterministic distributed transformations

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build deterministic distributed transformations and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Distributed transformations must be deterministic: same logical input/state should produce same logical output. Avoid order-dependent behavior without an explicit sort rule.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Derive latest shipment state using event_time + deterministic tie-breaker, not whichever row happens to appear last.

BUILD THIS ARTIFACT

models/SQL/dbt + lineage + row-count controls

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-04-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-04/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-05.

P03-05 - Join reference data and validate cardinality

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete join reference data and validate cardinality and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Reference joins can fan out too. Prove one-row-per-hub dimension before joining and reconcile shipment/event grain after.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Duplicate H1 in reference data and show how row count can grow even if every fact row still looks valid.

BUILD THIS ARTIFACT

p03_phase_artifact + evidence note

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-05-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-05/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-06.

P03-06 - Reason about partitions, shuffle and skew

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete reason about partitions, shuffle and skew and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Partition/shuffle/skew decisions need evidence. Start with group counts and physical plan; optimize one bottleneck without changing correctness.

TINY PATTERN - NOT YOUR FINAL SOLUTION

If HOT owns 80% of rows, repartitioning by that key will not magically balance work. Measure distribution first.

BUILD THIS ARTIFACT

Spark job/notebook + explain plan + partition/performance evidence

VALIDATE

Compare output to small trusted control result; inspect plan/partitions; justify performance choice without changing correctness.

CHANGED / FAILURE TEST

Supply skewed/unbalanced input and require correctness-preserving diagnosis; optimization is secondary to invariant safety.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-06-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-06/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-07.

P03-07 - Write Parquet with justified partition strategy

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete write parquet with justified partition strategy and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Parquet partitioning is a storage/query decision. Do not partition by near-unique shipment_id; prefer bounded columns aligned to pruning/backfill.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Compare event_date partitioning against shipment_id using file/partition counts and likely filter patterns.

BUILD THIS ARTIFACT

Spark job/notebook + explain plan + partition/performance evidence

VALIDATE

Compare output to small trusted control result; inspect plan/partitions; justify performance choice without changing correctness.

CHANGED / FAILURE TEST

Supply skewed/unbalanced input and require correctness-preserving diagnosis; optimization is secondary to invariant safety.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-07-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-07/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-08.

P03-08 - Apply Delta-style incremental merge/idempotency

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete apply delta-style incremental merge/idempotency and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Delta-style merge correctness depends on match key and event ordering. Rerunning the same update must not create new logical state.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Apply S03 update + S07 insert twice. Latest-state row count/result should be stable on exact rerun.

BUILD THIS ARTIFACT

state ledger/watermark + rerun/backfill evidence

VALIDATE

Run same input twice and prove trusted logical result is stable; then add a changed/new record and prove only intended state changes.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-08-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-08/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-09.

P03-09 - Define Bronze/Silver/Gold lakehouse contracts

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete define bronze/silver/gold lakehouse contracts and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Bronze/Silver/Gold are responsibility boundaries, not folder colors. State grain, trust and allowed transformations at each layer.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Bronze received events; Silver validated/latest-state as designed; Gold business metric. Reconcile transitions.

BUILD THIS ARTIFACT

source_contracts.md

VALIDATE

Read outputs back, verify schema/counts/keys, and prove partition/schema-evolution behavior on a changed input.

CHANGED / FAILURE TEST

Add/remove/change one field according to supplied compatibility rule; quarantine/fail safely rather than silently corrupt.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-09-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-09/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-10.

P03-10 - Handle schema evolution and supplied failure cases

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete handle schema evolution and supplied failure cases and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Schema change handling must distinguish additive compatible change from breaking change. Save failure before repair.

TINY PATTERN - NOT YOUR FINAL SOLUTION

carrier_code addition can be preserved/merged intentionally; missing shipment_id is breaking and must quarantine/fail.

BUILD THIS ARTIFACT

first-failure evidence + root_cause.md + repair + fresh retry evidence

VALIDATE

Save first failing evidence before repair; identify root cause; apply smallest safe repair; fresh retry must pass all prior controls.

CHANGED / FAILURE TEST

MANDATORY SUPPLIED FAILURE: capture first failure, diagnose, repair, fresh retry; do not overwrite original evidence.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-10-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-10/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-11.

P03-11 - Inspect plans/performance and prove equivalent results

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete inspect plans/performance and prove equivalent results and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Performance work requires equivalent-result proof. If a tuning change alters business output, it is not an optimization.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Save plan/file/distribution evidence before and after one change, then prove row/key/measure equality.

BUILD THIS ARTIFACT

Spark job/notebook + explain plan + partition/performance evidence

VALIDATE

Compare output to small trusted control result; inspect plan/partitions; justify performance choice without changing correctness.

CHANGED / FAILURE TEST

PHASE CHECK P03-11: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-11-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-11/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-12.

P03-12 - Map the design to Fabric/cloud deployment

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete map the design to fabric/cloud deployment and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Cloud translation maps responsibilities, not button clicks. Label identity, storage, compute, orchestration, secrets and monitoring.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Map local Bronze/Silver/Gold to Fabric Lakehouse/OneLake + notebook/pipeline only where responsibility matches.

BUILD THIS ARTIFACT

Fabric deployment design + connection/access/secret boundary notes

VALIDATE

Map identities/connections/access/secret boundaries; prove no plaintext secret or broad unsafe access is required.

CHANGED / FAILURE TEST

PHASE CHECK P03-12: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-12-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-12/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: P03-13.

P03-13 - Clean rerun, reconcile outputs and defend trade-offs

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete clean rerun, reconcile outputs and defend trade-offs and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Clean rerun plus 10x defense ties correctness to scale. State which claims are measured locally and which remain hypotheses.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Reproduce trusted controls, then name the first bottleneck you would measure at 10x and why.

BUILD THIS ARTIFACT

business_questions.md + assumptions.md

VALIDATE

Run same input twice and prove trusted logical result is stable; then add a changed/new record and prove only intended state changes.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-P03-13-EVIDENCE. Save:

03_PROJECT_WORKSPACES/P03_SCALABLE_LAKEHOUSE_PIPELINE/evidence/P03-13/attempt.json. P03 rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-01.